📘 Covariance

✅ 1. What is Covariance?

Covariance is a statistical measure that describes the **direction of the linear relationship** between two variables.

- **Positive covariance**: When one variable increases, the other tends to increase.
- **Negative covariance**: When one variable increases, the other tends to decrease.
- **Zero covariance**: No linear relationship.

✅ 2. Covariance Formula

a. Population Covariance

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^{N} (X_i - \mu_X)(Y_i - \mu_Y)$$

b. Sample Covariance

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$$

Where:

- X_i, Y_i are the individual data points,
- $\bar{X}, \bar{Y}$ are the sample means,
- N is the population size,
- n is the sample size.

✅ 3. Numerical Example

Suppose we have two variables:

Person	X (Hours Studied)	Y (Test Score)
A	2	65
B	4	70
C	6	75
D	8	80
E	10	85

We'll compute **sample covariance**.

✅ Step-by-Step Computation

Step 1: Compute the Means

$$\bar{X} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$$
$$\bar{Y} = \frac{65+70+75+80+85}{5} = \frac{375}{5} = 75$$

Step 2: Compute Each Product of Deviations

$$(X_i - \bar{X})(Y_i - \bar{Y}) =$$

Person	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	Product
A	2	65	-4	-10	40
B	4	70	-2	-5	10
C	6	75	0	0	0
D	8	80	2	5	10
E	10	85	4	10	40

Step 3: Sum of Products

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 40 + 10 + 0 + 10 + 40 = 100$$

Step 4: Apply the Sample Covariance Formula

$$\text{Cov}(X,Y) = \frac{1}{n-1}\sum (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{5-1} \times 100 = \frac{100}{4} = 25$$

✅ 4. Interpretation

- **Cov(X, Y) = 25** → Positive value means **hours studied and test scores increase together**, indicating a **positive linear relationship**.
- However, the **magnitude of covariance** is **not standardized**—it depends on the units of X and Y. To get a standardized measure, we use **correlation**.

✅ 5. Covariance Matrix

If you have multiple variables, the **covariance matrix** summarizes covariances between all variable pairs.

For variables X_1, X_2, X_3:

$$\text{Cov}(X) = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \text{Cov}(X_1, X_3) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \text{Cov}(X_2, X_3) \\ \text{Cov}(X_3, X_1) & \text{Cov}(X_3, X_2) & \text{Cov}(X_3, X_3) \end{bmatrix}$$

Each diagonal element is a **variance**, and off-diagonals are **covariances**.

✅ Summary

Concept	Description
Covariance	Measure of **direction** of relationship between two variables
Positive Value	Both variables increase together
Negative Value	One increases while the other decreases
Zero	No linear relationship
Formula	$\text{Cov}(X, Y) = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$
Standardized Form	**Correlation**

Correlation

✓ What is Correlation?

Correlation is a **statistical measure** that expresses the **strength and direction** of a linear relationship between two variables. It's commonly used in statistics, data analysis, and machine learning.

The most common correlation measure is the **Pearson correlation coefficient**, denoted by r. It is a **unitless number** between −1and +1.

✓ Formula for Pearson Correlation Coefficient

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Where:

- x_i, y_i are the individual sample values
- $\bar{x}, \bar{y}$ are the sample means of x and y
- n is the number of observations

Interpretation of *r*:

r value	Strength	Direction
+1	Perfect correlation	Positive
0.7 to 0.9	Strong	Positive
0.4 to 0.6	Moderate	Positive
0.1 to 0.3	Weak	Positive
0	No correlation	—
-0.1 to -0.3	Weak	Negative
-0.4 to -0.6	Moderate	Negative
-0.7 to -0.9	Strong	Negative
-1	Perfect correlation	Negative

✅ **Dataset**

Person	X (Hours Studied)	Y (Test Score)
A	2	65
B	4	70
C	6	75
D	8	80
E	10	85

✅ Step-by-Step Calculations

Step 1: Means

$$\bar{X} = \frac{2+4+6+8+10}{5} = 6, \quad \bar{Y} = \frac{65+70+75+80+85}{5} = 75$$

Step 2: Compute Components

Let's compute the following for each observation:

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})^2$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
2	65	-4	-10	16	100	40
4	70	-2	-5	4	25	10
6	75	0	0	0	0	0
8	80	2	5	4	25	10
10	85	4	10	16	100	40

Step 3: Compute Sums

$$\sum (X_i - \bar{X})^2 = 16 + 4 + 0 + 4 + 16 = 40$$

$$\sum (Y_i - \bar{Y})^2 = 100 + 25 + 0 + 25 + 100 = 250$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 40 + 10 + 0 + 10 + 40 = 100$$

Step 4: Plug into Formula

$$r = \frac{100}{\sqrt{40} \cdot \sqrt{250}} = \frac{100}{\sqrt{10000}} = \frac{100}{100} = 1$$

✅ Final Answer

$$\boxed{r = 1.0}$$

✅ Interpretation

Result	Meaning
r =1	Perfect **positive linear** relationship — when hours studied increases, test scores increase in a perfectly consistent way.

✅ Summary Table

Statistic	Value
Mean of X	6
Mean of Y	75
Covariance	25
Standard Deviation X	$\sqrt{40/4} = \sqrt{10} \approx 3.162$
Standard Deviation Y	$\sqrt{250/4} = \sqrt{62.5} \approx 7.906$
Correlation r	**1.0**

✅ Real-World Dataset Example: House Prices Dataset

Suppose you are working with a dataset of house sales, containing features like:

House Size (sqft)	Number of Bedrooms	Distance to City Center (km)	House Price ($)
1400	3	5	300,000
1800	4	3	450,000
1200	2	10	200,000
...	...	...	...

□ Use of Correlation

Let's say we want to understand **which features affect house price the most**.

📌 Step 1: Calculate Correlation Coefficients

Feature	Correlation with Price
House Size	+0.85
Number of Bedrooms	+0.65
Distance to City Center	–0.75

✅ Interpretation:

- **House Size** and **Price** have a strong positive correlation → as size increases, price tends to increase.
- **Distance to City Center** has a strong negative correlation → houses farther from the city tend to cost less.
- **Number of Bedrooms** also has a moderate positive correlation, but not as strong as size.

🔄 Applications in Data Science:

1. **Feature Selection**:
   - ✓ Drop redundant or irrelevant features (e.g., if two variables are highly correlated, pick one to avoid multicollinearity).
2. **EDA (Exploratory Data Analysis)**:
   - ✓ Understand trends, relationships, and dependencies between variables.
3. **Model Building**:
   - ✓ Helps in choosing linear vs. non-linear models.
   - ✓ Avoids including strongly correlated independent variables (to prevent overfitting or unstable coefficients).
4. **Anomaly Detection**:
   - ✓ If known relationships break down (e.g., a very large house with a surprisingly low price), it could be an error or an anomaly.

📉 Visual Insight: Correlation Heatmap

To summarize correlation among many variables, we often use a **correlation matrix** visualized as a **heatmap**.

```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Assuming df is your DataFrame
corr = df.corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

📚 **Summary**

Use of Correlation in Data Science	Purpose
Feature selection	Remove redundant inputs
Understanding relationships	Know how variables influence each other
Preprocessing for modeling	Handle multicollinearity
Improving interpretability of models	Know what drives your predictions
EDA and visualization	Spot patterns and trends early

Homework:

Feature	Correlation (r) with Price
size_sqft	+0.88 ✅ Strong positive
num_bedrooms	+0.52 ➕ Moderate positive
distance_city_center	–0.77 ❌ Strong negative
age_of_home	–0.22 ➖ Weak negative

Now interpret it.

🔥 What is a Heatmap?

A **heatmap** is a visual tool that represents **numerical values** (like correlations) using **color gradients** instead of plain numbers.

In data science, a heatmap is commonly used to display the **correlation matrix** of a dataset so that you can **see patterns** and **relationships between features at a glance**.

🏠 Hypothetical Housing Dataset

Let's recall the features in our dataset:

Feature	Description
size_sqft	Size of the house in square feet
num_bedrooms	Number of bedrooms
distance_city_center	Distance of the house from city center (km)
age_of_home	Age of the home in years
price	Sale price of the house

🎯 Goal of Heatmap

We want to understand:

- Which features are **strongly correlated** (either positively or negatively)
- Which features are **independent**
- Detect potential **multicollinearity**
- Decide which features may be **important predictors** of price

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 Step-by-Step: Correlation Matrix

Here's the **correlation matrix** we assumed (values are hypothetical):

	size_sqft	**num_bedrooms**	**distance_city_center**	**age_of_home**	**price**
size_sqft	1.00	0.82	–0.65	–0.30	0.88
num_bedrooms	0.82	1.00	–0.50	–0.25	0.52
distance_city_center	–0.65	–0.50	1.00	0.20	–0.77
age_of_home	–0.30	–0.25	0.20	1.00	–0.22
price	0.88	0.52	–0.77	–0.22	1.00

🎨 Color-Coded Heatmap (Conceptually)

Instead of numbers, a heatmap uses **colors to represent the strength and direction** of correlation.

Let's describe what the heatmap **colors** tell us:

Correlation Value	Color Meaning
1.0	Dark Blue (Perfect Positive)
0.7 to 0.99	Blue (Strong Positive)
0.3 to 0.7	Light Blue (Moderate Positive)
0.0 to 0.3	White (Weak/No Correlation)
–0.3 to 0.0	Yellow (Weak Negative)
–0.7 to –0.3	Orange (Moderate Negative)
–1.0 to –0.7	Red (Strong Negative)

📊 Heatmap Matrix (Textual Format)

Here's a **text-based simulation** of the heatmap:

	size_sqft	num_bedrooms	distance_city_center	age_of_home	price
size_sqft	1.00	0.82	–0.65	–0.30	0.88
num_bedrooms	0.82	1.00	–0.50	–0.25	0.52
distance_city_center	–0.65	–0.50	1.00	0.20	–0.77
age_of_home	–0.30	–0.25	0.20	1.00	–0.22
price	0.88	0.52	–0.77	–0.22	1.00

🔍 Interpretation of the Heatmap

1. size_sqft and price → ● 0.88

- Strong positive correlation
- Larger homes tend to be more expensive

2. distance_city_center and price → ● –0.77

- Strong negative correlation
- Houses far from city are cheaper

3. size_sqft and num_bedrooms → ● 0.82

- These are highly related (more bedrooms → likely bigger home)
- May lead to **multicollinearity** in regression models

4. age_of_home and all variables → Weak or moderate negative

- Slightly older homes tend to be cheaper, but effect is small

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 How Heatmaps Help in Data Science

Use Case	How Heatmap Helps
Exploratory Data Analysis	Spot patterns quickly without reading raw numbers
Feature Selection	Identify redundant features (e.g., size_sqft vs bedrooms)
Detect Multicollinearity	Avoid unstable regression models
Model Insight	Know which features are likely to affect prediction target

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✅ Want to Code This Heatmap?

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Example DataFrame
data = {
   'size_sqft': [1400, 1800, 1200, 1500, 1600],
   'num_bedrooms': [3, 4, 2, 3, 3],
   'distance_city_center': [5, 3, 10, 7, 4],
   'age_of_home': [10, 5, 20, 15, 8],
   'price': [300000, 450000, 200000, 350000, 400000]
}
df = pd.DataFrame(data)

# Generate correlation matrix
corr_matrix = df.corr()

# Plot heatmap
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap - Housing Data")
plt.show()
```